

3D Modeling & Design

1. Basic Requirements

The following are supplementary requirements for 3D Modeling & Design.

Applications

- Web browser (Chrome, Edge, or Safari)
- LocalSend
- Onshape FTC parts library — <https://ftconshape.com/> (maintained by FIRST and PTC; email FIRST@ptc.com to be added to the shared folder)
- FTC Insert Tool parts library plugin — <https://cad.onshape.com/appstore/apps/Design%20&%20Documentation/6515cfb91574253b1b96a6ba> (install from the Onshape App Store: [Subscribe](#) → [Get for Free](#))

Online Accounts

- CycleZLab — <https://www.cyclezlab.com/> (free, application required)
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2. Environment Setup

The following environment must be configured before you begin. If you have any questions, please consult an administrator.

Web Browser

Bookmark the following additional websites:

Name	URL
goBILDA	https://www.gobilda.com/
REV Robotics	https://www.revrobotics.com/
CycleZLab	https://www.cyclezlab.com/

3. Modeling Essentials

Part Design and Assembly

When performing 3D modeling in Onshape, pay special attention to the following:

- **Dimensional accuracy:** ensure all dimensions in the model match the physical parts
- **Assembly mates:** set up part-to-part mates (Mates and Mate Connectors) properly
- **Interference check:** run interference checks after assembly
- **Part naming:** name parts and assemblies according to unified conventions

Common FTC Parts Libraries

Commonly used FTC parts suppliers and resource platforms:

- **goBILDA:** provides a complete range of FTC structural components
- **REV Robotics:** provides control system modules and structural components

- **CycleZLab:** a FIRST robotics community platform (archive of CAD, code, and build logs)

Design Standards

- Use metric units (mm)
- Leave appropriate clearance for all 3D-printed parts (0.2–0.3 mm recommended)
- Export formats: STEP files for machining, STL files for 3D printing, and 3MF files for slicing